RECALLING Article 28(b) of the Convention on the International Maritime Organization con cerning the functions of the Committee, RECALLING ALSO resolution A.886(21) on Procedure for the adoption of, and amendments to, performance standards and technical specificatio ns, by which the Assembly resolved that the functions of adopting performance standards and technical specifications, as well as --- URL: https://wwwcdn.imo.org/localresources/ en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf the functions o f adopting performance standards and technical specifications, as well as amendments the reto, shall be performed by the Maritime Safety Committee on behalf of the Organization, TAKING INTO ACCOUNT the amendments to the International Convention for the Safety of Lif e at Sea, 1974 ("the Convention"), adopted by resolution MSC.496(105), NOTING, in partic ular, regulation IV/7.1.4 of the Convention, which requires every ship to be provided wi th a receiver or receivers capable of receiving maritime safety information and search a nd rescue related information throughout the entire voyage in w hich the ship is engaged , NOTING ALSO the works carried out by the International Telecommunication Union on a di gital navigational data (NAVDAT) system and the allocated frequencies on MF and HF for i ts use, NOTING FURTHER the Guidance for the reception of maritime safety information and search --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResol utions/MSCResolutions/MSC.569(109).pdf use, NOTING FURTHER the Guidance for the receptio n of maritime safety information and search and rescue related information as required i n the Global Maritime Distress and Safety System (GMDSS) (MSC.1/Circ.1645), which identi fies related information broadcast services and the equipment which should be installed on board ships to meet the requirements of chapter IV of the Convention, RECOGNIZING tha t further growth in information promulgated to ships is constrained by the capacity of t he International NAVTEX Service and HF NBDP broadcasts, and that MF and HF digital NAVDA T broadcasts may be used in the GMDSS, HAVING CONSIDERED, at its 109th session, the reco mmendation made by the Sub-Committee on Navigation, Communications and Search and Rescue at its eleventh session, 1 ADOPTS the Performance standards for the reception of maritim e safety information and search and rescue related information by MF and HF digital navi gational data (NAVDAT) --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre /IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf and search and rescue related inf ormation by MF and HF digital navigational data (NAVDAT) system, set out in the annex to the present resolution; 2 RECOMMENDS Governments to ensure that NAVDAT receiver equipmen t conforms to performance standards not inferior to those specified in the annex to the present resolution.

RESOLUTION MSC.569(109) (adopted on 6 December 2024) PERFORMANCE STANDARDS FOR THE RECEP TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 2 I:\MSC\10 9\22\MSC 109-22-Add.1.docx ANNEX PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAF ETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIO NAL DATA (NAVDAT) SYSTEM 1 INTRODUCTION 1.1 The NAVDAT receiver equipment should be capa ble of receiving maritime safety information (MSI) and search and rescue (SAR) related i nformation transmitted by NAVDAT --- URL: https://wwwcdn.imo.org/localresources/en/Knowl edgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf information (MSI) and s earch and rescue (SAR) related information transmitted by NAVDAT system on MF and HF fre quencies, and may be used to meet the requirements of SOLAS regulation IV/7.1.4.

1.2 In addition to meeting the requirements of the Radio Regulations, the NAVDAT receive r equipment should comply with the most recent versions of Recommendations ITU-R M.2010 and ITU-R M.2058 for NAVDAT in the MF and HF bands, respectively.

1.3 The NAVDAT receiver equipment should comply also with the requirements set out in re solutions A.694(17) , MSC.191(79), as amended by resolution MSC.466(101), and MSC.508(10 5), and the following performance standards.

2 GENERAL 2.1 The NAVDAT system allows the broadcast of messages in the form of digital files providing texts, pictures or any other data.

This broadcast can be done on the two international frequencies: 500 kHz in the MF band and 4 226 kHz in the HF band, and also on a ll other --- URL: https://wwwcdn.imo.org/loc alresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf fre quencies: 500 kHz in the MF band and 4 226 kHz in the HF band, and also on a ll other ma ritime radio frequencies assigned by the ITU for NAVDAT usage (see paragraph 4.2.1).

2.2 The NAVDAT receiver equipment should comprise a radio receiver, an appropriate anten na, a signal processor with non -volatile memory, a human -machine interface, data inter

faces (see paragraph 9) and: .1 an integrated display; or .2 a connection to external eq
uipment with a display (e.g.

an integrated navigation system).

2.3 The NAVDAT receiver could either be a stand -alone equipment or combined with anothe
r equipment.

2.4 The NAVDAT receiver equipment should be provided with an antenna capable of receivin
g the entire maritime radio frequencies from, at minimum, 400 kHz to 30 MHz.

2.5 The NAVDAT receiver equipment should provide facilities to automatically update the
ship's position and the time at which the position was determined from a suitable electr
onic --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolut
ions/MSCResolutions/MSC.569(109).pdf ship's position and the time at which the position
was determined from a suitable electronic position-fixing aid which may be an integral p
art of the equipment.

For equipment which does not have an integral position -fixing aid, such facilities shou
ld include a suitable interface conforming to the appropriate international standards.1
2.6 When the NAVDAT receiver equipment has a dedicated antenna, it is recommended that t
he antenna be equipped with two outputs for sharing with another MF/HF receiver.

1 Refer to IEC 61162.

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TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND
HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 3 I:\MSC\10
9\22\MSC 109-22-Add.1.docx 3 CONTROLS AND INDICATORS 3.1 Subject messages2 excluded by t
he operator from display should be readily available.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.569(109).pdf 3 CONTROLS AND INDICATORS 3.1 Subject messages2 excluded
by the operator from display should be readily available.

3.2 When a message is received correctly or with errors, the equipment may be able to em
it a corresponding short audio notification.

In addition, a distinctive alert should be built into the receiver for messages with dis
tress and urgency priority.

3.3 The NAVDAT receiver equipment should be capable of displaying communication paramete
rs, including received signal strength indication (RSSI), signal noise ratio (SNR) and b
it error rate (BER).

3.4 Each time a NAVDAT message is received, the NAVDAT receiver equipment should display
, in plain text, the priority, subject message, coast station identity and number of the
message.

3.5 The NAVDAT receiver equipment should be able to configure parameters of data interfa
ces for communication with other ship equipment.

3.6 The stored tables of the NAVDAT receiver equipment should be able to be updated ---
URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCR
esolutions/MSC.569(109).pdf interfaces for communication with other ship equipment.

3.6 The stored tables of the NAVDAT receiver equipment should be able to be updated by u
sing either a data interface or reception of stored tables update messages.

3.7 Human-machine interface, including the presentation of alerts, for the NAVDAT receiv
er equipment should be in accordance with the guidelines developed by the Organization.3
4 RECEIVER 4.1 Frequency management 4.1.1 The NAVDAT receiver should be capable of recei
ving transmissions on MF (500 kHz) and HF (4 226 kHz) channels simultaneously.

4.1.2 The NAVDAT receiver should also be able to receive, via a scanning function, on at
least one (or more) regional frequency allocated to NAVDAT in the MF and HF maritime ban
ds.

4.1.3 The NAVDAT receiver should demodulate signal(s) received on the MF channel 500 kHz
and the HF channel 4 226 kHz simultaneously.

4.1.4 The demodulated signal(s) received by scanning can be decoded simultaneously or --
- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MS
CResolutions/MSC.569(109).pdf 500 kHz and the HF channel 4 226 kHz simultaneously.

4.1.4 The demodulated signal(s) received by scanning can be decoded simultaneously or no
n-simultaneously.

4.2 Scan function 4.2.1 To allow reception of transmissions on national or regional freq uencies assigned to NAVDAT, the receiver should use a scan function on the following mar itime frequency bands: .1 the MF band from 415 kHz to 526.5 kHz (except 500 kHz); 2 See the list of subject message codes in the most recent version of Recommendation ITU -R M. 2010 or ITU-R M.2058.

3 Refer to Guideline on software quality assurance and human -centred design for e-navig ation (MSC.1/Circ.1512).

RESOLUTION MSC.569(109) (adopted on 6 December 2024) PERFORMANCE STANDARDS FOR THE RECEP TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 4 I:\MSC\10 9\22\MSC 109-22-Add.1.docx --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCe ntre/IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf BY MF AND HF DIGITAL NAVIGATI ONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 4 I:\MSC\109\22\MSC 109-22-Add .1.docx .2 the channels assigned to NAVDAT in appendix 15 of the Radio Regulations: 6 33 7.5 kHz, 8 443 kHz, 12 663.5 kHz, 16 909.5 kHz and 22 450.5 kHz (except 4 226 kHz); and .3 the frequency bands assigned to wideband digital transmissions in appendix 17 of the Radio Regulations in the bands 4, 6, 8, 12, 16 and 22 MHz.

4.2.2 The signals received on the frequencies selected for scanning can be decoded simul taneously or non-simultaneously according to the resources of the NAVDAT receiver at tha t moment.

4.2.3 The transmitted pre -signal4 should allow the NAVDAT receiver to detect the transm ission and tune in to the frequency, measure its SNR and identify the transmitting NAVDA T coast station, including, based on its location, the associated NAVAREA/METAREA.

4.3 The receiver sensitivity should be better than – 95dBm in a bandwidth of 10 kHz, wit h --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolution s/MSCResolutions/MSC.569(109).pdf 4.3 The receiver sensitivity should be better than – 9 5dBm in a bandwidth of 10 kHz, with a BER better than 10-4 after error correction.

4.4 The NAVDAT receiver equipment should comply with the latest appropriate electromagne tic compatibility (EMC) standards (e.g.

resolutions A.694(17) and A.813(19), and IEC 60945).

4.5 The NAVDAT receiver equipment should be able to receive any NAVDAT message and ident ify the priority level and the type of that message.

4.6 For identification purposes, the NAVDAT receiver equipment should use the Maritime M obile Service Identity (MMSI) of the ship (see also paragraph 11).

4.7 The NAVDAT receiver equipment should be able to select automatically whether to rece ive subsequent messages according to the following broadcast modes: .1 General broadcast : Broadcasting messages to all ships.

.2 Selective broadcast: Broadcasting messages to a group of ships or ships within a spec ific navigational area.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.569(109).pdf .2 Selective broadcast: Broadcasting messages to a group of ships or ships within a specific navigational area.

.3 Dedicated broadcast: Addressing messages to one ship, using MMSI.

4.8 The NAVDAT receiver equipment should be able to select automatically demodulation mo de, error correction decoding and message decoding methods according to the received mod ulation information stream (MIS).

4.9 NAVDAT message transmission is based on the transmission of data packets.

The equipment should be able to correct received messages using forward error correction and by using repeatedly sent data packets.

4.10 The NAVDAT receiver equipment should have a built -in real -time clock that is auto matically calibrated by a suitable electronic position-fixing aid (or clock reference th rough an onboard data network).

4 To ensure proper operation of the receiver scan function, the transmitters of active n ational or regional --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/In dexofIMOResolutions/MSCResolutions/MSC.569(109).pdf an onboard data network).

4 To ensure proper operation of the receiver scan function, the transmitters of active n ational or regional NAVDAT coast stations will broadcast a pre -signal before the NAVDAT frames.

This pre -signal is a known data repeated eight times for a total duration of 3.2 second
s.

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TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND
HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 5 I:\MSC\10
9\22\MSC 109-22-Add.1.docx 5 DISPLAY DEVICE 5.1 The display device should be able to dis
play different types of NAVDAT messages.

5.2 The design and size of the integrated display device should be such that displayed i
nformation is easily read under all conditions by observers at normal working distance a
nd viewing angles.

An interface to an external display device or appropriate navigat ional --- URL: https:/
/wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/M
SC.569(109).pdf viewing angles.

An interface to an external display device or appropriate navigat ional equipment should
be provided.

5.3 On the display, the following requirements should be met: .1 an indication of newly
received and unsuppressed messages should be immediately displayed until acknowledged or
for 24 hours after their receipt; and .2 newly received unsuppressed messages should als
o be displayed.

5.4 When displaying received messages, a clear indication of the end of any message shou
ld be given.

5.5 The display resolution should be at least 640*480 pixels.

5.6 The list of messages received on 500 kHz and 4 226 kHz can be displayed in different
areas of the display screen at the same time.

5.7 In the list of received messages, the latest message is displayed on the first line
with its number, contents, date and time.

6 STORAGE 6.1 Non-volatile memory 6.1.1 The NAVDAT receiver equipment should keep all in
formation concerning NAVDAT --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeC
entre/IndexofIMOResolutions/MSCResolutions/MSC.569(109).pdf 6 STORAGE 6.1 Non-volatile m
emory 6.1.1 The NAVDAT receiver equipment should keep all information concerning NAVDAT
coast stations in a non-volatile memory.

This information should include name of the stations, their geographical positions, thei
r numbering allocated by the NAVDAT coordination procedure, the frequency(s) used as wel
l as the different slots that can be used by the stations.

This information will be regularly updated upon receipt of stored tables update messages
.

6.1.2 For each frequency provided, it should be possible to record at least 100 messages
in the non-volatile memory provided.

It should not be possible for the user to erase messages from the memory.

When the memory is full, the oldest message should be replace d by new messages.

6.1.3 The user should be able to mark the individual messages for permanent retention.

These messages can occupy up to 25% of available memory and should not be overwritten by
--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.569(109).pdf These messages can occupy up to 25% of available memory
and should not be overwritten by new messages.

When no longer needed, the user should be able to delete the tag on the se messages, whi
ch can be overwritten normally.

6.1.4 The NAVDAT receiver equipment should be able to detect duplicate messages and disc
ard repeated messages.

6.1.5 A memory capacity of not less than 1 gigabyte should be provided to store, as a mi
nimum, the time, transmitter identity, subject and content of received messages.

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TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND
HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 6 I:\MSC\10
9\22\MSC 109-22-Add.1.docx 6.1.6 After between 60 and 72 hours, a message should automat
ically be erased from the memory storage.

If the number of received messages exceeds the capacity of the storage, the --- URL: htt

ps://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutio
ns/MSC.569(109).pdf memory storage.

If the number of received messages exceeds the capacity of the storage, the oldest messa
ge should be erased.

6.1.7 Only messages satisfactorily received (i.e.

BER is better than 10-4) should be stored.

6.2 Programmable control memories 6.2.1 Information identifying the transmitter service
area and the code of each subject message in programmable memory should not be erased by
interruptions in the power supply of less than 24 hours.

6.2.2 The NAVDAT receiver equipment should be able to display, delete and query stored m
essages, and be able to output messages manually or automatically to appropriate ship eq
uipment (e.g.

ECDIS).

7 ALERT 7.1 Upon receipt of new search and rescue related information messages with dist
ress or urgency priority, the NAVDAT receiver equipment should give an alert, including
information for decision-making.

7.2 The position information contained in the messages with distress and urgency --- URL
: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCReso
lutions/MSC.569(109).pdf for decision-making.

7.2 The position information contained in the messages with distress and urgency priorit
ies should be automatically transmitted to other navigation equipment (e.g.

ECDIS, ENC plotter).

7.3 The NAVDAT receiver equipment should comply with the Performance standards for bridg
e alert management (resolution MSC.302(87)).

8 TEST FACILITIES The NAVDAT receiver equipment should be provided with a facility to te
st the radio receiver, the display and the non-volatile memory and to display the result
s of the test.

9 DATA INTERFACES 9.1 The NAVDAT receiver equipment should include at least one interfac
e for the transfer of all messages to other navigation or communication equipment.

9.2 All interfaces provided for communication with other navigation or communication equ
ipment should comply with the relevant international standards.5 9.3 The NAVDAT receiver
equipment may include a standard printer interface.

10 POWER SUPPLY --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexo
fIMOResolutions/MSCResolutions/MSC.569(109).pdf 9.3 The NAVDAT receiver equipment may in
clude a standard printer interface.

10 POWER SUPPLY 10.1 The NAVDAT receiver equipment should normally be powered from the s
hip's main or emergency sources of electrical energy.

In addition, it should be possible to operate the equipment and all equipment necessary
for its normal functioning from the reserve so urce or sources of energy in accordance w
ith SOLAS regulation IV/13.

5 Refer to IEC 61162.

RESOLUTION MSC.569(109) (adopted on 6 December 2024) PERFORMANCE STANDARDS FOR THE RECEP
TION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND
HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM MSC 109/22/Add.1 Annex 12, page 7 I:\MSC\10
9\22\MSC 109-22-Add.1.docx 10.2 The NAVDAT receiver equipment should continue to operate
without any loss of software parameters or received messages stored in the memory in the
presence of power supply variations normally to be expected in a ship.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.569(109).pdf software parameters or received messages stored in the m
emory in the presence of power supply variations normally to be expected in a ship.

11 RECEIVER IDENTIFICATION It should be possible to configure the NAVDAT receiver equipm
ent with: .1 the identity (MMSI) of the ship (according to Recommendation ITU-R M.585);
or .2 the group identity (MMSI) (according to Recommendation ITU-R M.585); or .3 additio
nal lists of identities (MMSIs).

12 STORED TABLES The NAVDAT receiver equipment should have the possibility of memorizing
some tables.

For example: .1 The list of coast stations with: Area; Country; Longitude; Latitude; Nam

e; Slots; and Frequency used.

This stored table is queried when the identities of a NAVDAT coast station are received and the complete parameters of this NAVDAT coast station should be displayed in plain te xt.

.2 The list of subject messages: Table with subject message 01 to 63 (see the most recen t versions of --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofI MOResolutions/MSCResolutions/MSC.569(109).pdf should be displayed in plain text.

.2 The list of subject messages: Table with subject message 01 to 63 (see the most recen t versions of Recommendations ITU-R M.2010 and ITU-R M.2058).

*** RESOLUTION MSC.569(109) (adopted on 6 December 2024) PERFORMANCE STANDARDS FOR THE R ECEPTION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SYSTEM RESOLUTION MSC.569(109) (adopted on 6 D ecember 2024) PERFORMANCE STANDARDS FOR THE RECEPTION OF MARITIME SAFETY INFORMATION AND SEARCH AND RESCUE RELATED INFORMATION BY MF AND HF DIGITAL NAVIGATIONAL DATA (NAVDAT) SY STEM --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolut ions/MSCResolutions/MSC.573(110).pdf RESOLUTION MSC.573(110) (adopted on 26 June 2025) A MENDMENTS TO THE INTERNATIONAL CODE OF SAFETY FOR HIGH-SPEED CRAFT, 1994 (1994 HSC CODE) RESOLUTION MSC.573(110) (adopted on 26 June 2025) AMENDMENTS TO THE INTERNATIONAL CODE O F SAFETY FOR HIGH-SPEED CRAFT, 1994 (1994 HSC CODE) THE MARITIME SAFETY COMMITTEE, RECAL LING Article 28(b) of the Convention on the International Maritime Organization concerni ng the functions of the Committee, NOTING resolution MSC.36(63), by which it adopted the International Code of Safety for High-Speed Craft ("the 1994 HSC Code"), which has becom e mandatory under chapter X of the International Convention for the Safety of Life at Se a, 1974 ("the Convention"), NOTING ALSO article VIII(b) and regulation X/1.